

SE322: Software Design and Architecture

Activity 5: Principles of Detailed Design

Objective: this activity, the students will learn how to analyze an existing software architecture or design to check if the code violates principles or not.

Expected Outcome: After completing this exercise, the students will be able to:

1. Analyze code to check if the code violates principles or not
2. Analyze class diagram to check if the diagram violates principles or not

Course Learning Outcome:

1. CLO 1: Recognize the principles, strategies, and process of software design and architecture.
2. CLO 2: Analyze an existing software architecture or design.

Question1: Consider the code

```
public class Shape {
    ShapeType type;
}
public class Square extends Shape
{
    public Square() { this.type = ShapeType.SQUARE; }
    double side;
}
public class Circle extends Shape
{
    public Circle() { this.type = ShapeType.CIRCLE; }
    double radius;
    Point center;
}
public class Server
{
    void drawSquare(Square square)
    {
        System.out.println("I'm drawing a square");
    }

    void drawCircle(Circle circle)
    {
        System.out.println("I'm drawing a circle");
    }

    void drawAllShapes(List<Shape> shapes)
    {
```

```

for (Shape shape : shapes)
{
    switch (shape.type)
    {
    case SQUARE:
        drawSquare((Square) shape);
        break;
    case CIRCLE:
        drawCircle((Circle) shape);
        break;
    }
}
}
}

```

1. Does this code violate the OCP principle? Why?
2. If it is violated, rewrite the code in a way to comply with the OCP Principle.

Question2: Banking Application example

Banking application supports two account types – “current” and “savings”. The bank now wants to offer a high interest-earning fixed-term deposit account to its customers.

1. Consider the class diagram does it violate the LSP principle if the bank create fixed-term deposit as new account? Why?
2. If it is violated, redraw the diagram in a way to comply with the LSP Principle.

